

SANTEE COOPER, SC, USA

WMO: 723107

Lat: 33.587N

Lon: 80.209W

Elev: 31

StdP: 100.95

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00371

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.1	(c) -2.1	(d) -17.2	(e) 0.8	(f) 6.1	(g) -12.7	(h) 1.3	(i) 5.2	(j) 8.0	(k) 10.5	(l) 7.2	(m) 10.9	(n) 0.8	(o) 0	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.3	(c) 35.2	(d) 24.9	(e) 33.8	(f) 24.5	(g) 32.6	(h) 24.2	(i) 27.8	(j) 30.8	(k) 27.2	(l) 29.8	(m) 26.3	(n) 29.1	(o) 2.9	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	23.2	29.3	26.3	21.8	28.2	25.9	21.2	27.8	89.0	31.8	85.5	29.2	82.0	29.1	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.6	(b) 5.9	(c) 5.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 7.6	(b) 5.9	(c) 5.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	8.1	11.0	13.4	17.8	22.3	25.9	27.3	26.3	24.0	18.4	12.2	10.1		
	DBStd	7.83	5.59	5.02	5.08	4.35	3.45	2.44	2.42	2.37	2.57	4.14	4.37	5.13		
	HDD10.0	264	104	44	24	2	0	0	0	0	0	1	24	64		
	HDD18.3	1265	319	209	166	61	8	0	0	1	1	51	191	259		
	CDD10.0	3221	46	72	130	237	382	477	537	504	418	261	88	68		
	CDD18.3	1182	2	4	14	46	132	227	279	247	169	52	6	4		
	CDH23.3	11011	11	36	130	412	1173	2223	2952	2282	1353	396	34	11		
	CDH26.7	4406	0	2	24	99	438	943	1375	958	487	77	2	0		
(14) Wind	WSAvg	1.8	1.7	1.9	2.0	2.3	2.1	1.9	1.9	1.4	1.5	1.5	1.6	1.6		
(15) Precipitation	PrecAvg	1298	93	96	91	81	93	146	149	172	113	92	73	90		
	PrecMax	1663	190	225	156	172	190	346	291	271	214	270	173	184		
	PrecMin	980	25	32	14	22	32	58	69	105	16	0	5	30		
	PrecStd	188	52	45	40	41	38	70	55	53	55	63	46	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	27.0	29.1	31.1	35.1	37.2	37.5	36.4	33.0	31.1	27.1	25.1		
		MCWB	17.0	16.4	18.2	21.6	22.7	24.2	25.5	24.9	23.4	22.6	19.6	19.0		
	2%	DB	21.4	24.1	27.2	29.2	32.5	34.8	35.9	33.9	32.3	28.1	23.8	22.5		
		MCWB	15.7	15.7	17.3	20.4	21.4	24.3	25.0	24.9	23.4	21.1	17.6	17.9		
	5%	DB	19.1	22.3	24.8	27.5	31.1	32.6	34.0	32.6	31.2	27.2	22.3	21.0		
		MCWB	14.0	15.0	16.6	19.1	21.4	23.4	25.2	24.9	23.2	20.7	17.1	18.0		
	10%	DB	17.3	20.1	22.3	26.1	29.7	32.0	32.6	31.4	29.9	25.9	20.2	18.7		
		MCWB	13.7	14.3	14.9	18.4	21.1	23.2	25.1	24.3	22.8	20.0	16.6	16.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	19.5	20.6	24.3	25.0	28.2	29.0	28.7	27.1	25.6	21.8	21.3		
		MCDB	22.0	23.3	23.2	28.3	29.2	32.8	33.3	31.7	28.5	26.9	24.4	22.8		
	2%	WB	17.4	17.9	19.5	22.8	24.0	26.7	28.0	27.7	26.0	23.6	20.4	20.2		
		MCDB	19.2	21.3	23.5	26.3	29.0	29.1	31.2	30.3	29.0	26.8	22.0	21.4		
	5%	WB	16.2	16.8	18.3	21.6	23.3	25.6	27.3	27.0	25.3	22.6	18.8	18.4		
		MCDB	18.6	20.5	22.9	25.3	28.1	28.6	30.0	29.5	28.8	25.4	21.0	19.9		
	10%	WB	14.3	15.3	17.1	20.4	22.5	24.9	26.5	25.9	24.3	21.6	17.2	16.8		
		MCDB	17.1	18.7	20.6	24.1	27.2	28.8	29.5	28.8	28.2	24.3	19.1	18.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	12.4	12.9	12.9	11.5	10.7	10.3	10.0	10.6	12.0	12.6	10.9		
		MCDBR	13.1	14.4	14.5	13.8	13.0	11.7	11.2	11.2	12.1	12.6	14.0	12.5		
	5% WB	MCWBR	8.6	9.2	6.6	6.5	6.1	4.9	4.0	4.2	5.5	7.1	9.1	9.0		
		MCWBR	11.1	12.7	12.5	11.4	11.2	10.1	9.6	9.5	11.0	9.9	10.9	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.329	0.364	0.408	0.475	0.507	0.551	0.520	0.436	0.392	0.344	0.319		
	taud		2.514	2.487	2.413	2.279	2.139	2.095	1.993	2.070	2.296	2.390	2.485	2.554		
	Ebn at Noon		895	923	912	881	822	792	755	772	827	840	858	877		
	Edh at Noon		84	96	111	133	155	162	178	162	124	105	86	77		
(44) All-Sky Solar Radiation	RadAvg		2.81	3.44	4.54	5.75	6.19	6.39	6.14	5.53	4.75	4.02	3.10	2.49		
	RadStd		0.15	0.35	0.36	0.39	0.47	0.45	0.39	0.29	0.41	0.48	0.24	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+192	+135

Nomenclature: See separate page